



DeepFilter: An ECG baseline wander removal filter using deep learning techniques

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ARTICLE INFO

Keywords:

ECG
Baseline wander
Deep learning
Noise filtering

ABSTRACT

According to the World Health Organization, around 36% of the annual deaths are associated with cardiovascular diseases and 90% of heart attacks are preventable. Electrocardiogram signal, acquired whether during exercise stress test or resting conditions, allows cardiovascular disease diagnosis. However, during the acquisition, there is a variety of noises that may damage the signal quality thereby compromising their diagnostic potential. The baseline wander is one of the most undesirable noises. In this work, we propose a novel algorithm for BLW noise filtering using deep learning techniques. The model performance was validated using the QT Database and the MIT-BIH Noise Stress Test Database from Physionet. In addition, several comparative experiments were performed against state-of-the-art methods using traditional filtering as well as deep learning techniques. The proposed approach yields the best results on four similarity metrics, namely: the sum of squared distance, maximum absolute square, percentage of root distance, and cosine similarity with 5.20 ± 7.96 au, 0.39 ± 0.28 au, 50.45 ± 29.60 au and, 0.89 ± 0.1 au, respectively. The source codes of the proposed model as well as the implementation of related techniques are freely available on Github.

1. Introduction

Cardiovascular diseases (CVDs) constitute one of the main leading cause of sudden cardiac death worldwide. According to the World Health Organization (WHO), 17.9 million deaths are attributed to CVDs every year, which represents 31% of the global deaths [1]. Moreover, 85% of these deaths are related with heart attack and strokes. Nonetheless, recent studies have shown that, with early and effective diagnosis, 90% of heart attacks could be preventable [2]. The electrocardiogram (ECG) is a well-established technique for the diagnosis of heart diseases. It consists of acquiring the electrical activity of the heart over time by using an external electrode attached to the skin. The ECG can estimate the physical heart condition and detect a wide variety of abnormalities such as arrhythmia, coronary artery block, among others. It can be acquired under different conditions such as resting, ambulatory electrocardiography, and during an exercise stress test [3]. The last two allow a better diagnosis but are strongly affected by different sources of noise. For instance, the 60/50 Hz power line noise, contact noise, patient-electrode motion artifacts, EMG noise and baseline drift [4]. The ambulatory electrocardiography, also known as Holter monitoring or ambulatory ECG, allows to record the electrical activity of the heart during daily activities. It usually takes

24 h or more to record. Furthermore, the likelihood of encountering heart events is higher compared with other acquisition approaches. On the other hand, the exercise stress test (also known as Treadmill test, exercise test, exercise cardiac stress test, and ECST) records the heart function, usually between 5–15 min, while the subject performs an effort in a bicycle or on an electric treadmill. Many aspects of the heart function can be checked including heart rate, breathing, blood pressure, and how tired the patient becomes when exercising. Despite the increased probability of a better heart event detection in ECG long-term recording, new challenges arise. For instance, both the ambulatory and the stress test ECGs are prone to be highly contaminated with noise given their acquisition conditions [5].

Amongst the variety of noises, the baseline wander (BLW), also known as baseline drift, is one of the most challenging. This low-frequency noise contains components ranging between 0.05 Hz and 3 Hz [5]. It represents a serious limitation during stress tests and ambulatory ECG. In a typical graph showing a contaminated ECG signal, it can be observed that the time axis seems to deviate down instead of being straight. In other words, this effect consists in a movement of the whole signal from its normal base. The BLW can be caused by misplaced electrodes (electrode–skin impedance), bad skin preparation, patient's

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<https://doi.org/10.1016/j.bspc.2021.102992>

Received 9 January 2021; Received in revised form 3 June 2021; Accepted 17 July 2021

Available online 28 July 2021

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movement and breathing, scarce contact between electrode cables, and a combination of the aforementioned factors. It negatively affects the heart events detection in automatic diagnosis algorithms, yielding a useless outcome in most of the cases. Consequently, removing this noise is crucial to guarantee a proper clinical assessment from ECG recordings.

Several solutions have been proposed to mitigate the BLW. For instance, classical digital filters [6], Wavelet transforms [7] and empirical mode decomposition [8], just to name a few. Romero et al. [9] presented a comparative study between many of these approaches. While some methods obtained acceptable results, the morphological deformation in the ECG signal is still an outstanding problem. Although the BLW noise filtering has been object of extensive research, it remains an open problem. Hence, we propose to revisit it with a renewed vision by leveraging artificial intelligence-driven filters.

This paper proposes a novel algorithm for BLW noise filtering using deep learning techniques. We used the QT Database and the MIT-BIH Noise Stress Test Database from Physionet to perform several experiments. Also, an ablation of different models was conducted to determine the importance of linear and non-linear activations, and the advantage brought by dilated convolutions. We hypothesize that by learning “smart filters” it is possible to discriminate between the desired ECG signal and the undesired noise. Multiple deep filters will learn how to properly filter small sections of the input ECG signal while conserving its morphology using a similarity loss function.

In the following section, we start by reviewing some related techniques proposed for BLW removal. Section 3 provides details on the databases used. Section 4 contains the pre-processing techniques and the proposed deep learning models. Results and discussion are presented in Section 5. Finally, Section 6 concludes the paper and identifies further research directions.

To ensure reproducibility, the source code of this work is freely available in GitHub [<https://github.com/fperdigon/DeepFilter>]. Also, for educational purposes, our codes can be executed on Google Colab, where free GPU interactive sessions using Jupyter notebooks are available. [[Colab link](#)].

2. Related works

During several years, BLW noise removal has been an active research topic. Indeed, different techniques have emerged as alternatives to solve this problem, namely filtering, cubic splines, and independent component analysis, to name a few. In [10], the authors implemented a two-stage cascade least mean square adaptive filter. The first stage used a high-pass notch filter while the second stage contained a correlated filter of adaptive impulse. In contrast, Wavelet-based approaches uses different Wavelet families and threshold combinations [7]. In a comparative study, Chaudhary et al. [11] showed that Wavelet Sym10 with a semi-soft threshold yielded the best results for BLW removing. On the other hand, the moving average technique originally proposed by Canan et al. [12], acts as a low pass filter and simultaneously leverages the temporal domain. The resulting signal after applying the filter is subtracted from the original signal.

Finite and infinite impulse response filters [13,14] are considered as a classical filtering approach. Both methods have been widely used for BLW noise mitigation. Finite impulse response (FIR) filters tend to be more stable than the infinite impulse response (IIR) variant. However, the latter are usually preferred since they obtain similar results while using significantly less computational resources. Cubic spline [15] is another technique used for BLW estimation. It consists on interpolating the PR segments of each beat. Subsequently, the resulting signal is subtracted from the original signal.

In [16], the authors proposed to use independent component analysis for BLW removal. This approach is based on applying different delays to the input signal, thereby generating a set of multiple channels. In that work, there were generated 60 channels from a single ECG signal

using versions with delays between 11 and 20 samples. These versions formed an input set called mixed element matrix. Then, the independent components corresponding to BLW were detected automatically using the kurtosis. Finally, the components that were detected as BLW were set to zero and the original matrix was recomposed to extract the original ECG.

The surveys presented in [17] and [9] contain a comprehensive overview of relevant BLW noise removal techniques. The comparative study conducted by Romero et al. [9] showed that FIR and IIR filters yielded the best results for BLW noise mitigation according to several signal distortion metrics. Recently, deep learning techniques have shown great potential for processing a variety of biomedical signals. Specifically, deep autoencoders has been applied for ECG denoising. In [18], the authors introduced a denoising autoencoder (DAE) based on a fully convolutional network (FCN) yielding state-of-the-art results. Similarly, in [19] was also used the DAE to enhance the ECG signal. Alternatively, the work presented in [20] used a transfer learning technique by pre-training the network with synthetic data and fine-tuning with real data.

3. Materials

The QT Database [21] from Physionet [22] was used in all the experiments performed in this work. It contains real ECG records representing a wide range of QRS and ST-T morphologies in all the possible variability. Furthermore, it is composed of 105 ECG signals with 15-min and two channels each one sampled at 250 Hz. In our experiments, we used all the records available in this dataset. In order to add real baseline drifts to the QT database, real BLW signals from the MIT-BIH Noise Stress Test Database (NSTDB) [4] were used. This database contains 30 min of 12 ECG recordings and 3 recordings of typical noise in stress tests at a sampling frequency of 360 Hz. These noises are baseline wander produced by the patient’s breathing, muscle artifacts and electrode motion artifacts. The ECG records were randomly corrupted with the noise present in the three noise channels. The noise records were recorded during physical stress tests, with electrodes placed on the limbs in positions where ECG cannot be acquired. In this paper, we only used records with BLW noise generated by respiration and electrode motion artifacts.

4. Methods

In this section we present our deep learning BLW filtering solution called *DeepFilter*. The proposed model is a fully convolutional architecture based on multipath modules. It receives an ECG signal contaminated with BLW. The goal is to output the ECG signal without noise. Fully convolutional networks (FCN) are widely used for tasks where the output dimensions must be the same as the input dimension. As application examples of FCN we can mention U-net for semantic segmentation [23], VoxelMorph for deformable medical image registration [24], and black and white image colorization [25] for image restoration.

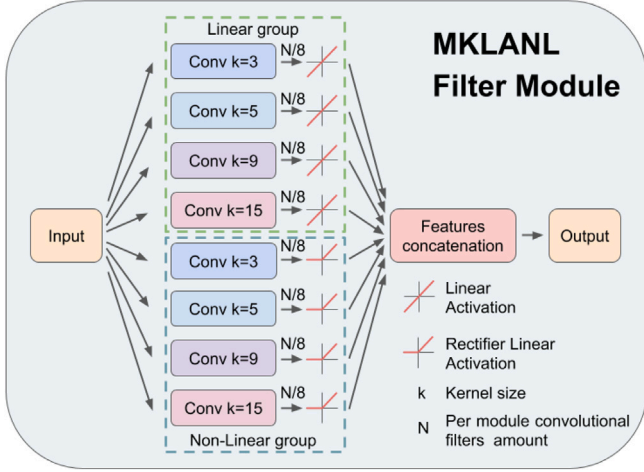
Our approach uses multipath modules, which place different convolutional layers at the same level and let the backpropagation algorithm to choose not only the weights but also the best path for the signal to pass through. Fig. 1 shows the proposed Multi-Kernel Linear And Non-Linear (MKLANL) filter module, which is inspired by the Inception module, originally introduced by Szegedy et al. [26]. In deep learning models, often the kernel size is one hyperparameter challenging to set. Therefore, typically a grid search or multiple empirical choices are performed to find an optimal value, which can be time-consuming. In contrast, we propose to employ multipath modules, inspired on the Inception architecture. Using this scheme, an optimal kernel will be learn at each level.

The proposed MKLANL filter module is composed by two internal groups: the linear group and the non-linear group. Each internal group

Table 1

Signals used for test set representing all internal data on the QT dataset.

MIT-BIH Arrhythmia Database	MIT-BIH ST Change Database	MIT-BIH Supraventricular Arrhythmia Database	MIT-BIH Normal Sinus Rhythm Database	European ST-T Database	Sudden death patients from BIH	MIT-BIH Long- Term ECG Database
sel123 sel233	sel302 sel307	sel820 sel853	sel16420 sel16795	sele0106 sele0121	sel32 sel49	sel14046 sel15814

**Fig. 1.** Proposed multi kernel linear and non-linear filter module.

contains four types of convolutional layers with kernels equal to (3, 5, 9, and 15) followed whether by a linear activation or a rectified linear unit (ReLU), depending on the group. The number of convolutional filters for each type is $N/8$, where N is a hyperparameter to control the total amount of filters per multipath module. Generally, N is referred as the amount of extracted features since the result of each convolution operation has associated a transformed output signal. The rationale behind having convolution with linear and non-linear activations is the same as having different convolutional kernels: let the model choose during training which path is better, and how much each of them will contribute to the output.

Generally, the effect of low-frequency signals is better appreciated when a large amount of consecutive samples is under analysis. That is the main reason why good low frequency filters have large windows. The equivalent procedure in convolutional networks would be increasing the kernel size, however, this has a negative effect on the computational load. Dilated convolutions are a smart and elegant solution to this problem, by using non-consecutive kernels [27]. This solution allows to increase the kernel receptive fields while the computational load remains the same (see Fig. 2).

For a one dimensional signal $x[i]$, the output $y[i]$ of a dilated convolution operation with a dilation rate r and a filter mask $w[s]$ is defined by the following formula:

$$y[i] = \sum_{s=1}^S x[i + r \cdot s] w[s] \quad (1)$$

Fig. 3 illustrates the proposed deep learning architecture. It is composed by six MKLANL filter modules arranged sequentially. The first two ((b) and (c)) have a total of 64 convolutional filters and are internally structured as shown in Fig. 1. In the second multipath module (c) the convolutional operations were configured with a dilation rate $r = 3$, enabling dilated convolutions. This same dilation rate was also set in the fourth and sixth modules. The number of extracted features decreases along the network, starting with 64 on modules one and two, then 32 features on modules three and four. Lastly, the modules five

and six contain 16 features each. The final step has one convolutional filter with kernel = 9, which conforms the output signal. Since ECG are bipolar signals, linear activation was used for this final step, thereby allowing the output to have either positive or negative values.

As previously mentioned, the model input (box (a) in Fig. 3) is an ECG signal contaminated with the BLW noise from the NSTDB. As ground-truth, we used the same ECG input signal but without BLW noise contamination. The goal is to learn the kernel values for BLW filtering with a low distortion through the iterative backpropagation process.

During the optimization process, the total loss function ($L_{filtering}$) combined both the Sum of Squared Distance (SSD) and the Maximum Absolute Distance (MAD), as shown in Eq. (2). The rationale behind this choice was to achieve a better representation for the task. The SSD term (left term in Eq. (2)) describes the signals difference along the sample dimension and does not minimize the effect of outlier values as averaged metrics such as Mean Squared Error (MSE) or the Signal to Noise Ratio (SNR). The MAD term (right term in Eq. (2)) computes the signal's difference maximum value. This term enforces the model to minimize the point-wise difference between ground-truth and predicted signals while the SSD term analyzes the whole signal. Since both losses have different value ranges, we used a balance term (λ). We empirically found that $\lambda = 50$ works well in our deep learning setup.

$$L_{filtering} = \sum_{m=1}^r (y_{true}(m) - y_{pred}(m))^2 + \lambda \times \max |y_{true}(m) - y_{pred}(m)|^r \quad (2)$$

where:

y_{true} : Original signal (ECG without noise)

y_{pred} : Predicted signal (ECG filtered by the model)

λ : Balance term

m : Sample variable

r : Maximum signal size

5. Experiments and results

5.1. Experimental design

For our experiments we designed an BLW removal benchmark where different methods can be evaluated using the same test set under the same conditions. Fig. 5 shows the data preprocessing pipeline. The 105 signals of the QT Database were over-sampled from 250 Hz to 360 Hz in order to match the NSTDB sampling frequency. Heartbeats were extracted using the specialist's annotations. We detected a small number of wrong annotations regarding the beat start/end. This causes two consecutive beats to be considered as one. To mitigate this error we discarded the beats greater than 512 samples (1422.22 ms). ECG and noise signals were acquired with a gain of 200 adu/mV (analog to digital unit per milliVolt). We considered 200 adu/mV equivalent to an arbitrary unit (au).

We divided the heartbeats belonging to 14 signals (see Table 1) to be used as test set, representing the 13% percent of the total signal amount. Two signals from each of the seven datasets were used to create the QT Database. This will ensure to have different pathologies on the test set, thereby giving a better idea of the generalization capability of the methods under evaluation.

The noise from the NSTDB, containing BLW (signal named *bw* on the database), was used to contaminate the ECG signals. Fig. 4 and

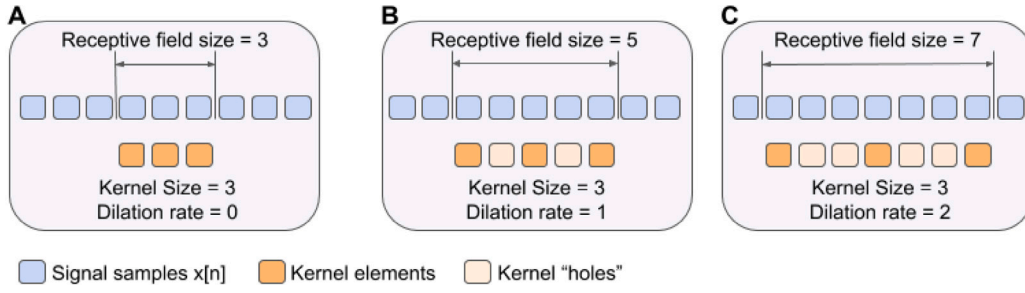


Fig. 2. Dilated convolutions. **A** is a dilated convolution with dilation rate $r = 0$, equivalent to the standard convolution. **B** has dilation rate $r = 1$ and **C** a dilation rate $r = 2$. All dilated convolution have kernel size equal to 3, the same number of parameters, and the same computational cost.

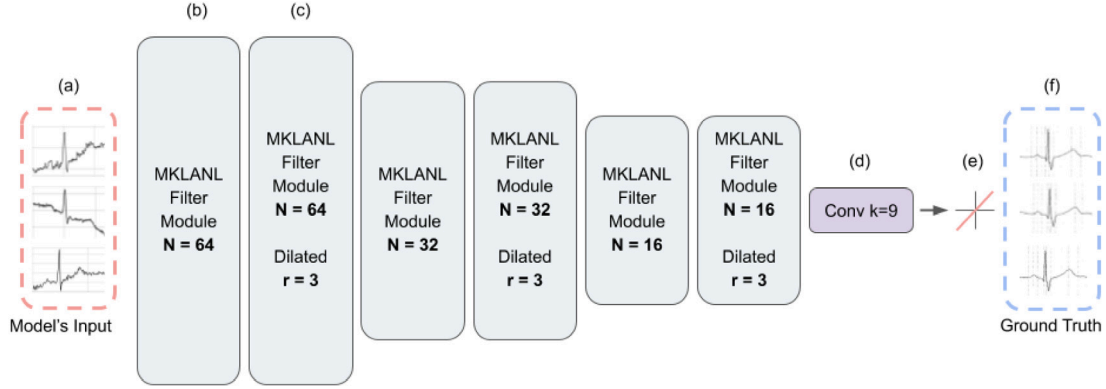


Fig. 3. Schematic representation of the proposed *DeepFilter* model. It is composed by a sequence of several MKLANL filter modules.

Table 2
Noise split used in our experiments.

Split name	Train noise	Test noise
Noise split v1	Channel 1_a	Channel 2_b
Noise split v2	Channel 2_a	Channel 1_b

Table 2 show the noise split used at training and testing stages. Each channel of the *bw* record was divided in half. Then two noise split schemes were created in a way that different noise channels were used for train and test. Also, the used selection avoids spurious correlation between channels by selecting noise chunks of different time intervals. Test results for the two noise channel configurations were grouped together and shown in Table 4.

Having a separate signal test set ensures the reliability of the results when using learning-based algorithms. This also prevents learning algorithms to have any sort of advantage when comparing with other classical methods.

To inject the noise we followed the same approach used by the NSTDB in [4], where the noise was randomly injected with values from 0.2 up to 2 times the ECG signal maximum peak value.

5.2. Implementation of comparative methods

We implemented state-of-the-art approaches to compare their performance with the proposed method. FIR and IIR filters were used in our experiments as a representation of non-learnable approaches. These classical filters obtained the best results on a comparative study performed by Romero et al. [9]. Also, we implemented two deep learning based state-of-the-art methods for ECG denoising. The first one based on deep recurrent neural networks (DRNN) [20] and the second one based on fully convolutional denoising autoencoder (FCN-DAE) [18]. Finally, in order to show the relevance of non-linear activations in deep models, the advantages of using the multi-branch architecture and

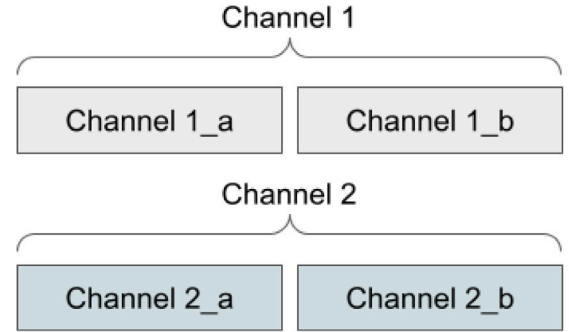


Fig. 4. NSTDB split.

the importance of dilated convolutions, we implemented three model variants for an ablation study.

FIR filter: The classical high-pass FIR digital filter presented in [13] is one of the methods that have reported high accuracy for BWL removing. Moreover, it has been considered as a robust solution for such a task [9]. The authors in [13] calculated this filter with 28 length Kaiser window and 51 coefficients. Nevertheless, in this work, this filter was redesigned taking into account the computational capabilities of modern hardware. The width transition from stop band to pass band 0.07 and an attenuation in the stop band of 30.5 dB. As a result, the design parameters were an order of 8079 and a beta of 2.18. In addition, it is important to highlight that the chosen cutoff frequency was 0.67 Hz, which has been proven a proper high-pass frequency for BWL removal according to the AHA [28].

IIR filter: Another remarkable classical filter that has stood out in state-of-the-art literature and experiments [9] is the IIR high pass filter proposed in [6]. Its main advantage is requiring less computational resources and at the same time being able to achieve high accuracy as its FIR filter counterpart. The design is composed by a Butterworth

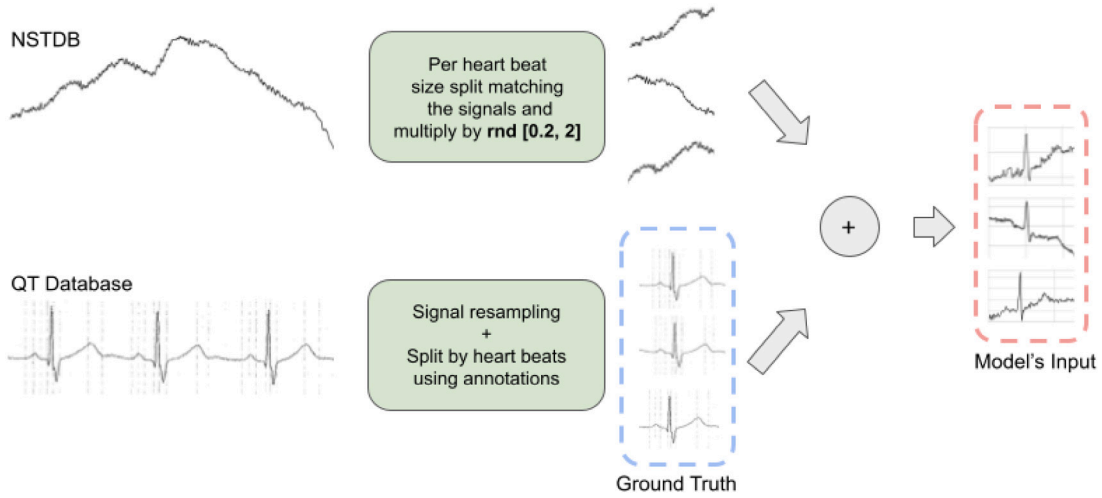


Fig. 5. Data preprocessing pipeline.

Table 3
Models considered for the ablation study.

Vanilla FCN linear	Vanilla FCN non-linear	Multibranch
64 Conv, $k = 9$, linear	64 Conv, $k = 9$, ReLU	64 MKLANL filter module
64 Conv, $k = 9$, linear	64 Conv, $k = 9$, ReLU	64 MKLANL filter module
32 Conv, $k = 9$, linear	32 Conv, $k = 9$, ReLU	32 MKLANL filter module
32 Conv, $k = 9$, linear	32 Conv, $k = 9$, ReLU	32 MKLANL filter module
16 Conv, $k = 9$, linear	16 Conv, $k = 9$, ReLU	16 MKLANL filter module
16 Conv, $k = 9$, linear	16 Conv, $k = 9$, ReLU	16 MKLANL filter module
1 Conv, $k = 9$, linear	1 Conv, $k = 9$, linear	1 Conv, $k = 9$, linear

filter with order 4. Moreover, in order to avoid the common phase delay in the signal output, it is crucial to notice that, bi-directional filters are used in both filters [29]. These filters achieve a zero-phase distortion by evaluating the input signal in both the forward and reverse directions. **Deep recurrent neural network (DRNN):** We also implemented the method based on recurrent networks proposed in [20]. As described on the original paper, we used a LSTM module with 64 units, the *return_sequence* parameter was set to *True* since this allows to return the full sequence, meaning that the output tensor will have the same shape as the input tensor. The LSTM was followed by two fully connected layers (also called dense layers) of 64 neurons each with ReLU activation. The final layer was composed of a dense layer with one neuron using linear activation. The loss function was the MSE as described in the original paper [20].

FCN-DAE: Denoising autoencoders allows learning architectures to automatically extract the most relevant features for the further reconstruction in presence of noise at the input by exploiting the generative capabilities of autoencoder models [30]. Chiang et al. [18] used a fully convolutional denoising autoencoder following the same principles as original autoencoders. The authors included batch normalization for better gradient flow and used the MSE as loss function.

Ablation study: We implemented three different models to assess the influence of different architecture variations. All models shared the same number of convolutional filters at the same level to provide a similar architectural root. They were tested using the proposed loss function and the same training protocol, which will be detailed in the next section. Table 3 shows the architecture used for the aforementioned models. The *Vanilla FCN Linear* model employed linear activation after the convolutional filters. In contrast, the *Vanilla FCN Non-Linear* used ReLU, a non-linear activation, in all its internal filters but the last one. Finally, the *Multibranch model* used the MKLANL filter module introduced in this work with dilation rate equal 0, meaning no dilation.

5.3. Training protocol

The training protocol explained in this section was applied to all deep learning based methods.

Train/Validation/Test split: As previously mentioned, we divided the heartbeats belonging to 14 signals to be used as a test set (two from each dataset used to conform the QT Database). This represents the 13% percent of the total signals amount. The heart beats from the remaining 91 signals were randomly shuffled and split to be used as train (70%) and validation (30%) sets. For learning based methods the same training set of signals + noise was used. Furthermore, the same testing set was used to evaluate all the methods.

Batch size: A batch size of 32 signals was used for training. This brings the advantage of the speed-up of matrix-matrix products making the training faster [31].

Optimization and learning rate scheduling: Adam optimization algorithm [32] was used with an initial learning rate of 10^{-3} . This learning rate was reduced by a factor of 2 after 2 epochs without improvements in the validation set. The minimum allowed learning rate was 10^{-10} .

Iterations: The initial number of epochs was 10^5 . Due to the early stopping criterion that was implemented, the final number of epochs was automatically determined during training. For the early stopping, the SSD metric over the validation set was monitored, and stopped after 10 epochs if no improvements were detected.

5.4. Evaluation metrics

According to different studies, distance-based metrics are the most suitable for the evaluation of signal similarity [33,34]. In order to evaluate the performance of methods for BLW removing having minimal distortion on the signal morphology we used the following metrics: Absolute maximum distance (MAD), sum of the square of the distances (SSD), percentage root-mean-square difference (PRD), and cosine similarity.

Maximum absolute distance (MAD): This metric was used in [34] in the context of compression of ECG signals. It is considered a well-known similarity metric to quantify the ECG quality after being processed. It measures the maximum absolute distance between the original and filtered signal. The formula is given by:

$$MAD(s_1, s_2) = \max |s_1(n) - s_2(n)| \quad 1 \leq n \leq N \quad (3)$$

where s_1 and s_2 are the original and filtered signals, respectively; n is the index of the current sample and N is the length of the signals. Both signals must have the same length. The same applies to the Sum of the

Table 4Evaluation metrics results for different methods. Values are mean $\pm$ std.

Method/Model	SSD (au)	MAD (au)	PRD (%)	Cosine Sim $\times 100$ (%)
FIR filter	47.464 $\pm$ 88.496	0.676 $\pm$ 0.572	66.158 $\pm$ 22.104	0.692 $\pm$ 0.216
IIR filter	35.645 $\pm$ 70.189	0.594 $\pm$ 0.532	61.125 $\pm$ 23.220	0.736 $\pm$ 0.208
DRNN	6.702 $\pm$ 11.316	0.480 $\pm$ 0.340	50.035 $\pm$ 24.802	0.878 $\pm$ 0.126
FCN-DAE	10.633 $\pm$ 14.417	0.598 $\pm$ 0.381	94.504 $\pm$ 70.717	0.770 $\pm$ 0.171
Vanilla L	13.623 $\pm$ 16.365	0.538 $\pm$ 0.298	83.997 $\pm$ 29.832	0.728 $\pm$ 0.137
Vanilla NL	7.810 $\pm$ 11.795	0.431 $\pm$ 0.303	63.294 $\pm$ 35.386	0.845 $\pm$ 0.121
Multibranch LANL	6.772 $\pm$ 10.553	0.403 $\pm$ 0.291	58.389 $\pm$ 30.311	0.864 $\pm$ 0.114
Multibranch LANLD (proposed)	5.203 $\pm$ 7.915	0.386 $\pm$ 0.283	50.445 $\pm$ 29.598	0.897 $\pm$ 0.097

Table 5SSD metric values obtained with different methods and noise amplitudes. Values are mean $\pm$ std.

Method/Model	0.2 < noise < 0.6	0.6 < noise < 1.0	1.0 < noise < 1.5	1.5 < noise < 2.0
FIR filter	7.162 $\pm$ 9.485	22.542 $\pm$ 30.795	51.381 $\pm$ 72.620	95.983 $\pm$ 134.621
IIR filter	5.615 $\pm$ 7.508	17.262 $\pm$ 25.049	38.334 $\pm$ 58.050	71.980 $\pm$ 107.880
DRNN	4.000 $\pm$ 7.437	4.932 $\pm$ 6.652	7.016 $\pm$ 10.971	10.031 $\pm$ 15.665
FCN-DAE	7.396 $\pm$ 8.032	8.638 $\pm$ 8.247	10.929 $\pm$ 12.883	14.586 $\pm$ 21.429
Vanilla L	8.346 $\pm$ 8.954	10.183 $\pm$ 8.220	14.083 $\pm$ 14.400	20.185 $\pm$ 23.905
Vanilla NL	4.356 $\pm$ 7.463	6.140 $\pm$ 7.126	8.489 $\pm$ 11.492	11.267 $\pm$ 16.273
Multibranch LANL	3.282 $\pm$ 5.020	5.002 $\pm$ 5.686	7.325 $\pm$ 9.769	10.476 $\pm$ 15.421
Multibranch LANLD (proposed)	2.401 $\pm$ 3.419	3.816 $\pm$ 4.771	5.717 $\pm$ 7.781	8.078 $\pm$ 11.104

Table 6MAD metric values obtained with different methods and noise amplitudes. Values are mean $\pm$ std.

Method/Model	0.2 < noise < 0.6	0.6 < noise < 1.0	1.0 < noise < 1.5	1.5 < noise < 2.0
FIR filter	0.280 $\pm$ 0.183	0.510 $\pm$ 0.337	0.759 $\pm$ 0.513	1.045 $\pm$ 0.713
IIR filter	0.248 $\pm$ 0.171	0.449 $\pm$ 0.322	0.666 $\pm$ 0.487	0.918 $\pm$ 0.676
DRNN	0.390 $\pm$ 0.286	0.432 $\pm$ 0.273	0.496 $\pm$ 0.330	0.576 $\pm$ 0.409
FCN-DAE	0.480 $\pm$ 0.278	0.532 $\pm$ 0.301	0.617 $\pm$ 0.369	0.727 $\pm$ 0.474
Vanilla L	0.476 $\pm$ 0.240	0.499 $\pm$ 0.245	0.544 $\pm$ 0.279	0.616 $\pm$ 0.371
Vanilla NL	0.319 $\pm$ 0.220	0.377 $\pm$ 0.224	0.453 $\pm$ 0.289	0.544 $\pm$ 0.381
Multibranch LANL	0.291 $\pm$ 0.196	0.346 $\pm$ 0.207	0.423 $\pm$ 0.274	0.521 $\pm$ 0.376
Multibranch LANLD (proposed)	0.293 $\pm$ 0.214	0.341 $\pm$ 0.222	0.401 $\pm$ 0.270	0.483 $\pm$ 0.351

Table 7PRD metric values obtained with different methods and noise amplitudes. Values are mean $\pm$ std.

Method/Model	0.2 < noise < 0.6	0.6 < noise < 1.0	1.0 < noise < 1.5	1.5 < noise < 2.0
FIR filter	42.673 $\pm$ 15.504	61.370 $\pm$ 18.213	73.491 $\pm$ 17.577	81.627 $\pm$ 15.250
IIR filter	38.653 $\pm$ 15.668	55.867 $\pm$ 19.482	67.972 $\pm$ 19.714	76.644 $\pm$ 18.290
DRNN	40.298 $\pm$ 19.983	45.582 $\pm$ 21.936	52.317 $\pm$ 24.336	59.312 $\pm$ 27.389
FCN-DAE	76.445 $\pm$ 41.696	86.350 $\pm$ 56.163	98.331 $\pm$ 68.010	112.012 $\pm$ 94.656
Vanilla L	78.794 $\pm$ 34.009	81.750 $\pm$ 31.087	85.538 $\pm$ 28.114	88.451 $\pm$ 25.793
Vanilla NL	46.274 $\pm$ 25.272	56.990 $\pm$ 29.432	67.685 $\pm$ 34.481	77.698 $\pm$ 40.459
Multibranch LANL	40.644 $\pm$ 16.625	51.148 $\pm$ 21.605	62.443 $\pm$ 28.551	74.496 $\pm$ 36.636
Multibranch LANLD (proposed)	34.026 $\pm$ 14.234	43.346 $\pm$ 19.925	53.777 $\pm$ 27.189	66.167 $\pm$ 38.040

Table 8Cosine similarity values obtained with different methods and noise amplitudes. Values are mean $\pm$ std.

Method/Model	0.2 < noise < 0.6	0.6 < noise < 1.0	1.0 < noise < 1.5	1.5 < noise < 2.0
FIR filter	0.891 $\pm$ 0.082	0.760 $\pm$ 0.149	0.637 $\pm$ 0.192	0.532 $\pm$ 0.209
IIR filter	0.910 $\pm$ 0.074	0.800 $\pm$ 0.143	0.690 $\pm$ 0.191	0.589 $\pm$ 0.217
DRNN	0.926 $\pm$ 0.070	0.906 $\pm$ 0.084	0.871 $\pm$ 0.122	0.824 $\pm$ 0.167
FCN-DAE	0.844 $\pm$ 0.110	0.807 $\pm$ 0.136	0.757 $\pm$ 0.167	0.692 $\pm$ 0.205
Vanilla L	0.802 $\pm$ 0.108	0.764 $\pm$ 0.113	0.711 $\pm$ 0.128	0.654 $\pm$ 0.145
Vanilla NL	0.916 $\pm$ 0.062	0.875 $\pm$ 0.086	0.829 $\pm$ 0.115	0.778 $\pm$ 0.146
Multibranch LANL	0.931 $\pm$ 0.048	0.896 $\pm$ 0.072	0.852 $\pm$ 0.105	0.796 $\pm$ 0.145
Multibranch LANLD (proposed)	0.948 $\pm$ 0.042	0.921 $\pm$ 0.064	0.888 $\pm$ 0.093	0.844 $\pm$ 0.124

square of the distances and the Percentage root-mean-square difference metrics.

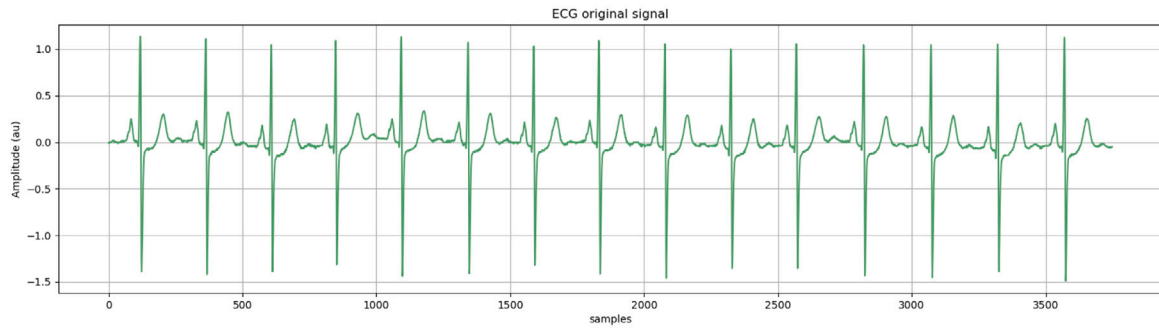
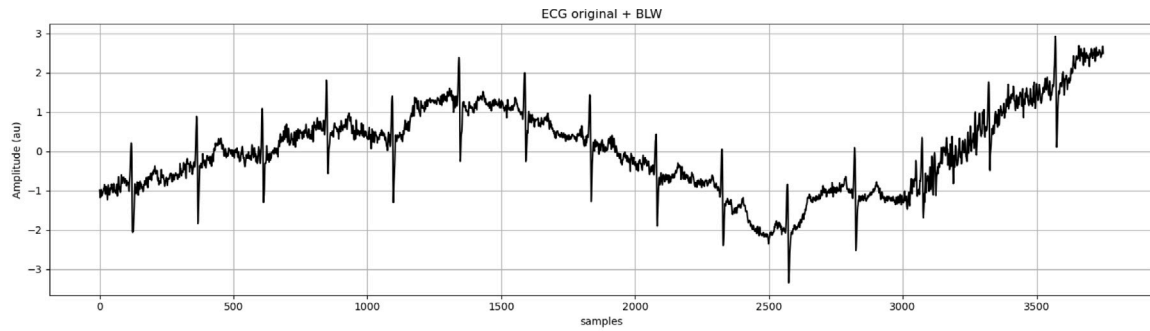
Sum of the square of the distances (SSD): This is another popular similarity metric [34], which measures the sum of squared distances between the original and the filtered signal. It provides an idea of how similar the signals are along their entire duration. The formula is given by:

$$SSD(s_1, s_2) = \sum_{n=1}^N (s_2(n) - s_1(n))^2 \quad 1 \leq n \leq N \quad (4)$$

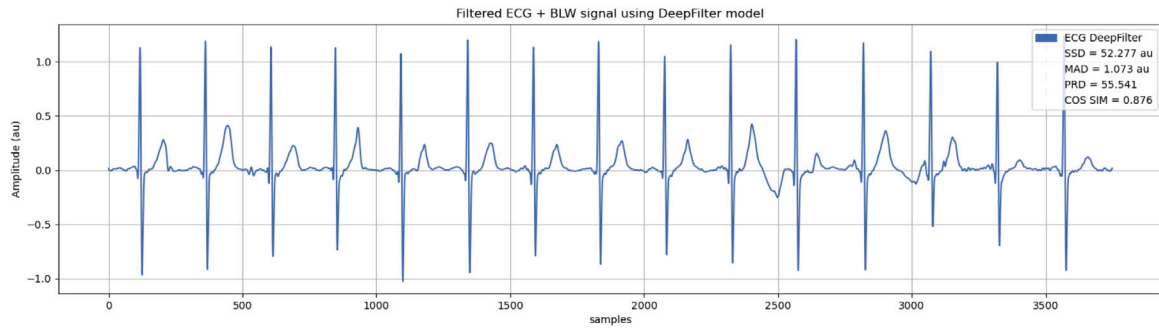
Percentage root-mean-square difference (PRD): This is a common similarity metric based on distance and its value is given in percentage. The objective is to obtain a smaller value, and consequently, the filtered signal is considered the best approximation of the original. The formula can be expressed as:

$$PRD(s_1, s_2) = \sqrt{\frac{\sum_{n=1}^N (s_2(n) - s_1(n))^2}{\sum_{n=1}^N (s_2(n) - \bar{s}_1)^2}} * 100\% \quad 1 \leq n \leq N \quad (5)$$

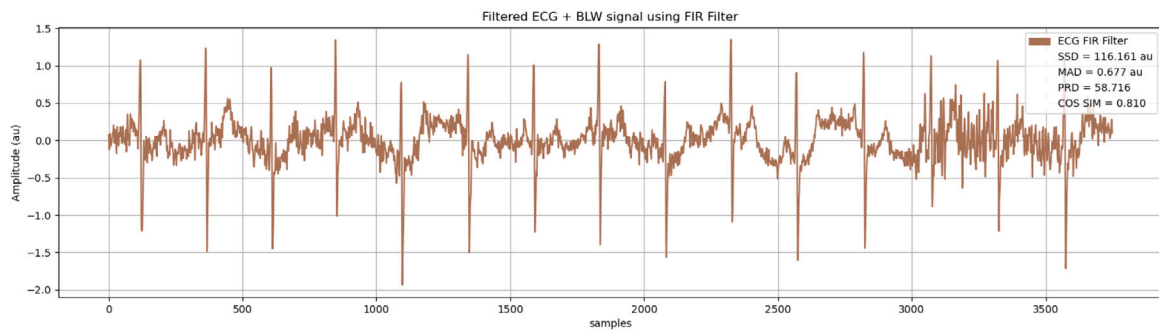
Cosine similarity: This is a measure of similarity between two vectors. It is a normalized bounded inner product by L2 norms. The cosine

(a) Portion of *sele0106* ECG signal.

(b) Original ECG + BLW noise from the NSTDB.

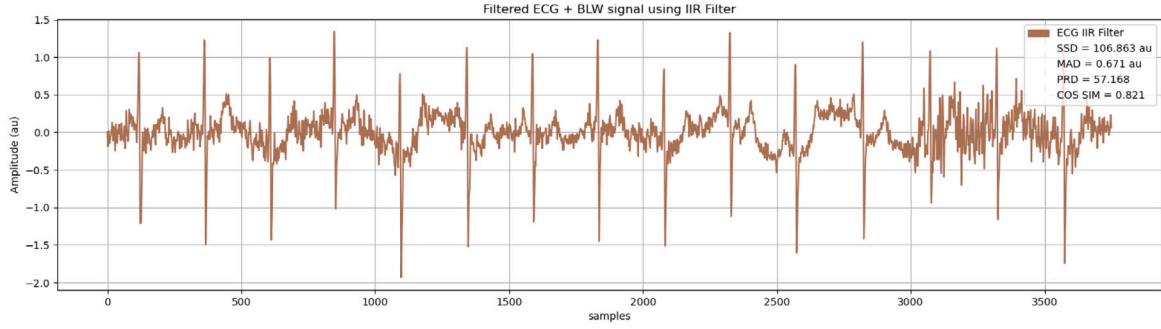


(c) ECG recovered using our approach.

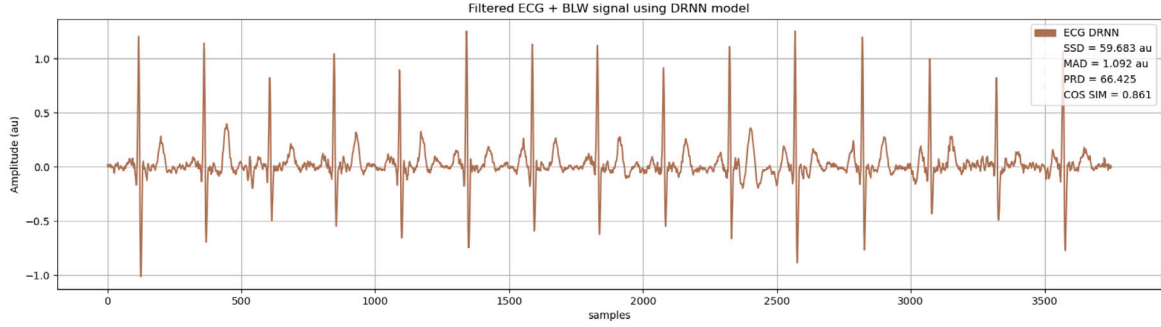


(d) ECG recovered using the FIR filter.

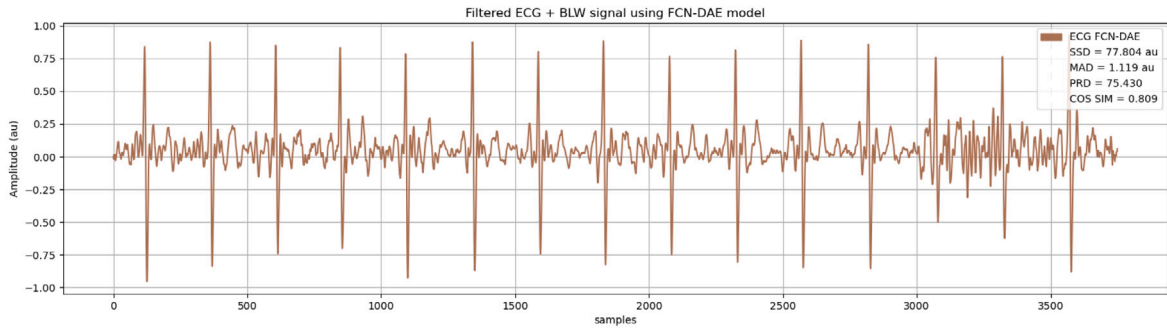
Fig. 6. Qualitative comparison of the recovered signal by the different methods.



(e) ECG recovered using the IIR filter.



(f) ECG recovered using the DRNN filter.



(g) ECG recovered using the FCN-DAE filter.

Fig. 6. (continued).

similarity comes because the normalized dot product by the Euclidean L2 normalization is the cosine of the angle between the points denoted by the vectors in the unit sphere [35]. The formula is given by:

$$\text{CosSim}(s_1, s_2) = \frac{\sum_{n=1}^N s_1(n)s_2(n)}{\sqrt{\sum_{n=1}^N s_1^2(n)}\sqrt{\sum_{n=1}^N s_2^2(n)}} = \frac{\langle \mathbf{s}_1, \mathbf{s}_2 \rangle}{\|\mathbf{s}_1\| \|\mathbf{s}_2\|} \quad 1 \leq n \leq N \quad (6)$$

where s_1 and s_2 are the original signal and the filtered one to be compared, n is the index of the current sample and N is the length of the signals. It is not necessary that both signals feature the same length, however in our particular use case signals length is always the same. It is important to notice that, as the cosine similarity gets closer to 1, the more similar are the two vectors (or the two signals).

5.5. Results and discussion

We now present a comparison between the proposed method, two state-of-the-art deep learning methods and two classical approaches that yielded good results on a previous comparative study. As explained in Section 5.2, FIR and IIR filters were implemented as described

by Van Alste et al. [13] and Kumar et al. [6], respectively. Also we implemented two recently proposed deep learning based methods, the first using DRNN [20] and the second based on denoising autoencoders [18].

Our experiments were aimed at investigating three main aspects: (1) the model that introduces less signal deformation, (2) the influence of linear and non-linear operations on the model performance, and (3) the influence of dilated convolutions. We determine the signal deformation using the following similarity metrics: Sum of Squared Distance (SSD), Maximum Absolute Distance, Percent of root-mean-square Distance (PRD), and Cosine Similarity. Statistical significance was calculated by applying a Wilcoxon signed-rank test using the Scipy library implementation. $P < 0.01$ was considered to indicate a statistically significant difference.

Table 4 presents the average values and the standard deviation yielded by the different methods/models (rows) and their respective metric values (columns). When comparing results we can observe that the proposed method obtained the best results on all the metrics. Moreover, it was closely followed by the DRNN [20] model. However, this slightly difference between the proposed method and DRNN was

found to be statistically significant ($p < 0.01$). FCN-DAE [18] model suffers from a well-known problem on auto-encoder architectures [31], which is related to fact that sometimes decoders are not able to recover fine details. This is due to the compressed representation from the input signal and could be solved by using skip connections like the U-Net architecture [23]. In Table 4 it can also be observed that the Vanilla FCN model including non-linearities outperforms the fully linear variant. Our intuition behind these results is that since models that feature non-linearities are able to learn more complex functions [36], the non-linear model is capable not only of performing a very strong filtering but signal reconstruction as well.

Tables 5 to 8 show the performance for each method when segmenting the test set by different noise amplitude considering the four similarity metrics. The overall performance is consistent with the results observed in Table 4. The proposed model outperforms the comparative methods in most of the tests (14 out of 16). As hypothesized, the model featuring dilated convolutions outperforms its counterpart. As was previously explained, dilated convolutions increase the field of view thereby allowing the model to better filter the low-frequency noise.

Fig. 6 shows qualitative and quantitative results obtained with our deep learning approach. The plot at the top (Fig. 6a) contains an ECG beat extracted from the signal *sele0106* of the QT Database (in green). Fig. 6b shows the *sele0106* signal + BLW extracted from the NSTDB. Fig. 6c displays the ECG signal recovered by our approach. Figs. 6d–6g present the results for the implemented comparative methods. Despite the differences between the original signal and the signal estimated by our approach, it can be observed that the morphology is highly preserved even when the signal was strongly contaminated by noise. For instance, it can be seen that, on the signal recovered by the proposed method, the amplitude of the QRS complex of the last 3 beats is highly altered. Also, the P wave in the last beat almost disappeared. Nonetheless, these are minor drawbacks if we consider the alternative of using the FIR and IIR filters, were the last 3 beats are almost indistinguishable from noise.

Another aspect that should be mentioned is the processing speed. Sometimes deep learning approaches tend to be discarded due to high processing time using CPU during deployment. While it is true that the training using CPU can take several hours, the time required for inference is minimum. This is due to the fact that our model is based on a fully convolutional scheme and also on optimizations such as dilated convolutions.

For instance, our model takes 48 ms to process one ECG beat while the implemented FIR filter takes 343 ms on the same CPU (Intel(R) Core(TM) i5-7300HQ CPU @ 2.50 GHz). Also, the proposed approach allows hardware implementation on an application-specific integrated circuit (ASIC) or a field-programmable gate array (FPGA) using a hardware description language (HDL) since convolutional filter modules can be easily parallelizable [37]. This will boost performance allowing the use on real-time applications such as professional ECG monitors.

6. Conclusion

In this work we presented a novel deep learning model for ECG baseline wander removal. The model performance was compared with related state-of-the-art methods, such as classical filtering, using similarity metrics. Furthermore, we conducted an ablation study to determine the importance of linear and non-linear activations and the use of dilated convolutions. The experimental results showed that the combination of linear and non-linear activations and dilated convolutions following a multi-kernel approach proved to be the best solution. The proposed model outperformed both state-of-the-art deep learning solutions and classical approaches. In summary, in this paper we have demonstrated the great performance of our method for eliminating

baseline wander noise. Further research should be focused on eliminating other noises that affect the ECG signal, such as muscle contraction (EMG), power line, and electrosurgical noise.

In order to maximize the reproducibility of this work, the source code will be freely available in GitHub [https://github.com/fperdigon/DeepFilter]. Therefore, it can be used as a benchmark for future research, streamlining the comparison with other models. Finally, for educational purposes, this work can be run on Google Colab with free GPU interactive sessions using Jupyter notebooks ([Google Colab link]).

Supplementary material

Code hosted in Github: [https://github.com/fperdigon/DeepFilter]

Run the codes using GPUs at Google Colab: [Google Colab link]

CRediT authorship contribution statement

Francisco P. Romero: Conceptualization, Methodology, Software, Formal analysis, Investigation, Visualization, Data curation, Writing – review & editing. **David C. Piñol:** Data curation, Writing – review & editing, Software, Validation. **Carlos R. Vázquez-Seisdedos:** Formal analysis, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

Thanks to Liset Vázquez Romaguera for fruitful discussions and document review.

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